Prodigy Anesthesia Quick Review

Cyclooxygenase Inhibitors

Do NSAIDs other than aspirin inhibit COX receptors? How do they differ?	Yes. NSAIDs such as indomethacin, phenylbutazone, and all other NSAIDs inhibit COX receptors. Unlike aspirin, however, they do not irreversibly bind to the receptor and the effect they have on platelets diminishes as the drug is cleared from the body and diffuses away from the receptor. *Barash PG, Cullen BF, Stoelting RK, Cahalan MK, Stock MC, Ortega R. Clinical Anesthesia. 7th ed. Philadelphia, PA: Lippincott Williams and Wilkins; 2013: 437.*
What COX-2 inhibitor is available for use in surgical patients?	Celecoxib (Celebrex) is currently the only COX-2 selective NSAID available. The others have been withdrawn from the market due to concerns for their cardiovascular side effects. Pain from surgery can be treated with this agent because it does not impair platelet aggregation associated with COX-1 receptor inhibition. *Nagelhout JJ, Plaus KL. Nurse Anesthesia. 5th ed. St. Louis, MO: Elsevier Saunders Company; 2014: 1249.*
How does the distribution of COX-1 and COX-2 receptors in the body differ?	COX-1 receptors are widespread throughout the entire body. COX-2 receptors are synthesized in response to inflammatory processes. *Nagelhout JJ, Plaus KL. Nurse Anesthesia. 5th ed. St. Louis, MO: Elsevier Saunders Company; 2014: 1249.*
How do the effects of COX-1 receptor binding by NSAIDs differ from that of COX-2 receptors?	COX-1 receptor inhibition by NSAIDs is responsible for the gastric irritation, decrease in renal blood flow, and platelet inhibition associated with nonselective NSAIDs. COX-2 receptor inhibition is responsible for the decrease in pain and inflammation. *Barash PG, Cullen BF, Stoelting RK, Cahalan MK, Stock MC, Ortega R. Clinical Anesthesia. 7th ed. Philadelphia, PA: Lippincott Williams and Wilkins; 2013: 437.*
How does inhibition of COX-1 and COX-2 receptors differ with respect to their effects on coagulation?	COX-1 receptor inhibition results in inhibition of thromboxane A2. By doing so, it disrupts thromboxane A2's ability to promote platelet aggregation. COX-2 receptor inhibition has been shown to reduce prostacyclin release from the vascular epithelium and can result in increased platelet aggregation. *Barash PG, Cullen BF, Stoelting RK, Cahalan MK, Stock MC, Ortega R. Clinical Anesthesia. 7th ed. Philadelphia, PA: Lippincott Williams and Wilkins; 2013: 437.*
How does aspirin promote bleeding?	Thromboxane A2 promotes platelet aggregation and causes vasoconstriction. Aspirin inhibits cyclooxygenase, which results in inhibition of the synthesis of thromboxane A2. *Barash PG, Cullen BF, Stoelting RK, Cahalan MK, Stock MC, Ortega R. Clinical Anesthesia. 7th ed. Philadelphia, PA: Lippincott Williams and Wilkins; 2013: 437.*

Copyright 2015 Prodigy Anesthesia, Inc. Unlawful to duplicate, distribute, or sell this document.